67%

ABERDARE TECHNICAL COLLEGE
INFORMATION COMMUNICATION TECHNOLOGY
UNIT CODE: 0612 451 07A

CAT 3

Total marks: 40 **Time: 1 hour 30 minutes**

~~26/12~~ 28/12

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Instructions to Candidates

- ✓ Answer ALL questions.
- ✓ Marks are shown in brackets at the end of each question or part.
- ✓ Write clearly and show your working where a calculation is required.
- ✓ Mobile phones and unauthorized materials are not allowed in the examination room.

QUESTIONS

1. Define network testing. **(2 marks)**

2. State two reasons why network testing is important. **(2 marks)**

3. Name the instrument used to measure current without cutting or disconnecting the wire, and state how a voltmeter is connected to the component being tested. **(2 marks)**

4. State the role of Npcap when using Wireshark on a Windows computer. **(2 marks)**

5. Define a VLAN. **(2 marks)**

Question 6 (6 marks)

(a) Define TTL and state what a router does when the TTL of a packet reaches 0. **(2 marks)**

(b) A Windows computer sends a packet with a starting TTL of 128. The packet passes through 9 routers before reaching its destination. Calculate the TTL on arrival and state whether the packet is dropped. **(2 marks)**

(c) Explain how the tracert command uses TTL to discover each router on the path to a destination. **(2 marks)**

Question 7 (6 marks)

A technician runs tracert to a web server and obtains the output below.

1	2 ms	1 ms	2 ms	192.168.1.1
2	9 ms	8 ms	9 ms	10.20.0.1
3	26 ms	25 ms	27 ms	41.89.10.5
4	180 ms	178 ms	182 ms	196.201.4.9
5	*	*	*	Request timed out.
6	190 ms	188 ms	191 ms	203.0.113.20

(a) Identify the hop where the delay increases sharply and state what this suggests about the network. **(2 marks)**

possible reasons for the asterisks (* * *) shown at hop 5, even though hop 6 replied. **(2 marks)**

(c) Explain why running ping alone would not have shown where the delay occurs. **(2 marks)**

Question 8 (6 marks)

(a) Describe the TCP three-way handshake as seen in Wireshark, stating who sends each packet and the SYN and ACK flag values in each step. **(4 marks)**

(b) Write the Wireshark display filter that would show only:

(i) the first packet (SYN) of each TCP connection request; **(1 mark)**

(ii) ICMP Echo Request packets. **(1 mark)**

Question 9 (6 marks)

A technician is commissioning a new office network and records the three findings below.

Finding	Tool used	Observation
(i)	Voltmeter	A wall socket reads 180 V AC. The expected supply is about 240 V AC.
(ii)	Cable tester	A newly crimped T568B patch cord shows LEDs 1 to 4 and 6 to 8 lighting in order, but LED 5 does not light.
(iii)	Wi-Fi analyser	The signal in the meeting room is measured at -82 dBm.

For each finding, interpret the result and recommend one suitable action. **(2 marks each)**

Question 10 (6 marks)

A school has a Finance department (PC1 and PC2 on ports Fa0/1 and Fa0/2) and a Human Resource department (PC3 and PC4 on ports Fa0/3 and Fa0/4) connected to one Cisco switch. The Finance network is 192.168.10.0/24 and the Human Resource network is 192.168.20.0/24. The school wants the two departments separated.

(a) Write the switch commands to create VLAN 10 named FINANCE and to assign port Fa0/1 to it as an access port. **(3 marks)**

(b) After the VLANs are configured, PC1 (192.168.10.10) cannot ping PC3 (192.168.20.10). Explain why, and describe how communication between the two VLANs can be enabled. **(3 marks)**

END OF QUESTION PAPER

1 Network testing is checking a network to ensure it works correctly and perform as expected.

2 To identify network Problems
To ensure reliable network Performance.

3 Instrument - Clampmeter
A Voltmeter is Connected in Parallel with the Component

4 It Captures network Packets
It allows Wireshark to access network traffic.

5 A VLAN is a Logical division of a network into Separate broadcast domains.

6 TTL means Time to Live
When TTL reaches 0, the Packet is dropped.

b Starting TTL = 128
Routers Passed = 9
TTL = 128 - 9 = 119

c) Traceroute Sends Packets with increasing TTL values. Each router that reduces TTL to zero responds, revealing its address.

7 a) The delay increases sharply at hop 4 where it rises at 180ms.

b. The router does not respond to traceroute Probe
c) A firewall blocks or filters the Probes.

d) Ping measures the response time to the destination only. It de

8 SYN - Client sends a Connection request to the Server.
SYN-ACK - Server responds with SYN and ACK flags.
ACK - Client sends an ACK to Complete the Connection. 8½

b) i) First TCP Packet (SYN):
tcp.flags.syn == 1 && tcp.flags.ack == 0 8

ii) ICMP Echo Request Packet:
icmp.type == 8 8

9 Interpretation - There may be a broken or incorrectly Connected wire on Pin 5
Action - Re-Crimp or replace the Patch Cord. 8½

ii) Interpretation - The Wi-Fi signal is weak 8½
Action - Move the Access Point Closer or install another Access Point.

10 a) Commands to Create VLAN 10 named FINANCE and assign Fa0/1 to it.
enable
Configure terminal
Vlan 10
Name FINANCE 8½
exit
interface fa0/1
Switchport mode access
Switchport access Vlan 10
end

b) Reason - VLANs separate broadcast domains, and devices in different VLANs cannot communicate directly through a layer 2 switch. 8

Solution - Configure inter-VLAN routing using a router or a Switch, with the correct IP addresses and default 8½